HRV 2022 Under Non-Homothetic CES Preferences.

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1 Non-Homothetic CES Preferences

This appendix summarizes the modifications to the benchmark HRV2022 environment when the household demand system is replaced by non-homothetic CES preferences over goods and services. All elements of the production side, the investment aggregator, sectoral prices, factor prices, and aggregate output remain exactly as in the benchmark model. The only changes arise on the household side and in the solution algorithm for consumption.

1.1 Non-homothetic CES aggregator

Let c_{gt} and c_{st} denote household consumption of goods and services. Aggregate consumption C_t is defined implicitly by

$$\omega_c^{1/\sigma}\left(\frac{c_{gt}}{C_t^{\eta_g}}\right)^{\frac{\sigma-1}{\sigma}}+(1-\omega_c)^{1/\sigma}\left(\frac{c_{st}}{C_t^{\eta_s}}\right)^{\frac{\sigma-1}{\sigma}}=1, \tag{1}$$

where σ is the elasticity of substitution between goods and services, $\omega_c \in (0,1)$ is the consumption preference weight, and η_g and η_s govern the income elasticities of demand.

The associated expenditure function is

$$E_t \equiv e(P_t, C_t) = \left[\omega_c P_{gt}^{1-\sigma} C_t^{(1-\sigma)\eta_g} + (1-\omega_c) P_{st}^{1-\sigma} C_t^{(1-\sigma)\eta_s}\right]^{\frac{1}{1-\sigma}}, \tag{2}$$

where E_t denotes the minimum expenditure required to attain the aggregate consumption index C_t at prices $P_t = (P_{gt}, P_{st})$.

1.2 Consumption demands and expenditure shares

By Shephard's lemma, sectoral consumption demands are

$$c_{gt} = \omega_c P_{gt}^{-\sigma} C_t^{(1-\sigma)\eta_g} E_t^{\sigma}, \tag{3}$$
$$c_{st} = (1-\omega_c) P_{st}^{-\sigma} C_t^{(1-\sigma)\eta_s} E_t^{\sigma}. \tag{4}$$

The corresponding consumption expenditure shares are

$$s_{gt}^C = \frac{P_{gt} c_{gt}}{E_t} = \omega_c P_{gt}^{1-\sigma} C_t^{(1-\sigma)\eta_g} E_t^{\sigma-1}, \tag{5}$$
$$s_{st}^C = \frac{P_{st} c_{st}}{E_t} = (1-\omega_c) P_{st}^{1-\sigma} C_t^{(1-\sigma)\eta_s} E_t^{\sigma-1}. \tag{6}$$

Taking the ratio of service to goods expenditure shares gives

$$\log \frac{s_{st}^C}{s_{gt}^C} = \log \frac{1-\omega_c}{\omega_c} + (1-\sigma) \log \frac{P_{st}}{P_{gt}} + (1-\sigma)(\eta_s - \eta_g) \log C_t.$$

Relative to the benchmark demand system, consumption structural transformation now depends on two forces. Relative price changes affect expenditure shares through the substitution elasticity σ, while growth in aggregate consumption affects expenditure shares through the difference $\eta_s - \eta_g$. In particular, if $\eta_s > \eta_g$, the service expenditure share rises with aggregate consumption.

1.3 Household problem and Euler equation

The representative household chooses a path for aggregate consumption C_t and capital accumulation to maximize

$$\max_{\{C_t, K_t\}_{t \geq 0}} \int_0^\infty e^{-\rho t} \frac{C_t^{1-\psi} - 1}{1-\psi} \, dt, \tag{7}$$

subject to

$$\dot{K}_t = Y_t - E_t - \delta K_t, \qquad E_t = e(P_t, C_t). \tag{8}$$

The current-value Hamiltonian is

$$\mathcal{H}_t = \frac{C_t^{1-\psi} - 1}{1-\psi} + \lambda_t \left[Y_t - E_t - \delta K_t \right]. \tag{9}$$

The first-order condition with respect to C_t is

$$C_t^{-\psi} = \lambda_t \frac{\partial E_t}{\partial C_t}. \tag{10}$$

The marginal expenditure cost of aggregate consumption is

$$\frac{\partial E_t}{\partial C_t} = E_t^{\sigma} \left[\omega_c \eta_g P_{gt}^{1-\sigma} C_t^{(1-\sigma)\eta_g - 1} + (1-\omega_c)\eta_s P_{st}^{1-\sigma} C_t^{(1-\sigma)\eta_s - 1} \right]. \tag{11}$$

Equivalently, using expenditure shares,

$$\frac{\partial E_t}{\partial C_t} = \frac{E_t}{C_t} \left[\eta_g s_{gt}^C + \eta_s s_{st}^C \right]. \tag{12}$$

The costate equation is

$$\dot{\lambda}_t = \lambda_t(\rho + \delta - R_t). \tag{13}$$

Combining (10) and (13) yields the Euler equation

$$R_t - \delta - \rho = \psi \hat{C}_t + \widehat{\left(\frac{\partial E_t}{\partial C_t} \right)}. \tag{14}$$

Relative to the benchmark HRV2022 Euler equation, the growth rate of aggregate consumption depends not only on the real return, but also on the evolution of the marginal expenditure cost of producing one additional unit of the non-homothetic consumption index.

1.4 Implications for the solution algorithm

The aggregate production side is unchanged. In particular,

$$Y_t = \mathcal{A}_{xt} K_t^{\theta}, \tag{15}$$

and factor prices remain

$$R_t = \theta\, \mathcal{A}_{xt} K_t^{\theta-1}, \tag{16}$$

$$W_t = (1-\theta)\, \mathcal{A}_{xt} K_t^{\theta}. \tag{17}$$

The aggregate balanced growth rate of capital and output is therefore unchanged:

$$\hat{K} = \hat{Y} = \frac{\hat{\mathcal{A}}_{xt}}{1-\theta} \equiv g_{\mathrm{ABGP}}. \tag{18}$$

Finding initial conditions. The initial per-capita capital stock k_0 is found by bisection. For each candidate k_0, three steps are executed.

1. The ABGP capital-accumulation condition pins down the initial per-capita consumption expenditure:

$$e_0 = \hat{\mathcal{A}}_{x0}\, k_0^{\theta} - \big(g_{\text{ABGP}} + \delta + n\big)\, k_0. \tag{19}$$

2. Given e_0, the composite consumption index C_0 and the preference weight ω_c are jointly calibrated by imposing the expenditure function together with the observed initial goods expenditure share $s_{g,0}^{\text{data}}$:

$$e(P_0,\, C_0) = e_0, \qquad s_g(P_0,\, C_0) = s_{g,0}^{\text{data}}. \tag{20}$$

   Because $\eta_g \neq \eta_s$, neither equation has a closed-form solution for (C_0, ω_c) individually, so the two conditions are solved numerically together.

3. The Euler equation is applied one step forward to obtain C_1, from which $E_1 = e(P_1, C_1)$ follows directly. The implied initial expenditure growth rate $\hat{E}_0 = \log(E_1/E_0)$ is then computed.

The bisection finds the unique k_0 satisfying

$$\hat{E}_0 = g_{\text{ABGP}}. \tag{21}$$

This is the *only* step in which the Euler equation is used.

Recursive simulation. Given $(k_t,\, C_t,\, E_t)$, each subsequent period is computed as follows.

1. Compute $\mathcal{A}_{xt}$, sectoral prices P_{gt} and P_{st}, aggregate output Y_t, and the rental rate R_t.

2. Impose the ABGP growth condition directly on consumption expenditure:

$$E_{t+1} = E_t\, e^{g_{\text{ABGP}}}. \tag{22}$$

3. Recover C_{t+1} by numerically inverting the NHCES expenditure function:

$$e(P_{t+1},\, C_{t+1}) = E_{t+1}. \tag{23}$$

4. Update the capital stock via the aggregate resource constraint and the capital-accumulation equation:

$$X_t = Y_t - E_t, \qquad \dot{K}_t = X_t - \delta K_t. \tag{24}$$

5. Recover sectoral consumption quantities and expenditure shares from the NHCES demand system (3)–(6).

The main computational differences relative to the benchmark model are therefore twofold. First, the Euler equation enters only in the k_0 bisection, not in the period-by-period dynamics. Second, the NHCES expenditure function must be numerically inverted at every step to recover C_{t+1} from the imposed E_{t+1}.

1.5 Properties of the Equivalent Variation Measure under Non-Homothetic CES Preferences

This subsection characterizes the equivalent variation measure under non-homothetic CES preferences. The construction follows the same two-stage logic as in the benchmark model. The key difference is that the household's optimal hypothetical consumption expenditure is no longer available in closed form. Instead, it is characterized by a scalar nonlinear equation in the aggregate consumption index.

Recall that aggregate consumption C is associated with the expenditure function

$$E(P_g, P_s, C) = \left[\omega_c P_g^{1-\sigma} C^{(1-\sigma)\eta_g} + (1-\omega_c) P_s^{1-\sigma} C^{(1-\sigma)\eta_s}\right]^{\frac{1}{1-\sigma}}. \tag{25}$$

The period utility function is

$$U(C) = \frac{C^{1-\psi} - 1}{1-\psi}, \tag{26}$$

and ν_t denotes the marginal value of investment evaluated at time t. The equivalent variation measure is defined as

$$\widehat{m}_{t,z,\tau} = f\left(u(m_\tau, P_{g,\tau}, P_{s,\tau}; \nu_t), P_{g,z}, P_{s,z}; \nu_t\right), \tag{27}$$

where the time-t preference representation is given by the quasilinear Bellman expression

$$U(C) + \nu_t x. \tag{28}$$

The equivalent variation measure is obtained by solving two consecutive problems.

First stage. At the first stage, the representative household uses time-t preferences, summarized by ν_t, to evaluate the utility attainable at time τ with income m_τ and prices $(P_{g,\tau}, P_{s,\tau})$. The household solves

$$\max_{\widetilde{C}_{t,\tau}, \widetilde{x}_{t,\tau}} \; U(\widetilde{C}_{t,\tau}) + \nu_t \widetilde{x}_{t,\tau} \tag{29}$$

subject to

$$m_\tau = E(P_{g,\tau}, P_{s,\tau}, \widetilde{C}_{t,\tau}) + \widetilde{x}_{t,\tau}. \tag{30}$$

Substituting the budget constraint into the objective function gives

$$\max_{\widetilde{C}_{t,\tau}} \; U(\widetilde{C}_{t,\tau}) + \nu_t \left[m_\tau - E(P_{g,\tau}, P_{s,\tau}, \widetilde{C}_{t,\tau}) \right]. \tag{31}$$

The first-order condition with respect to $\widetilde{C}_{t,\tau}$ is

$$U'(\widetilde{C}_{t,\tau}) = \nu_t \frac{\partial E(P_{g,\tau}, P_{s,\tau}, \widetilde{C}_{t,\tau})}{\partial C}. \tag{32}$$

Since $U'(C) = C^{-\psi}$, this becomes

$$\widetilde{C}_{t,\tau}^{-\psi} = \nu_t \frac{\partial E(P_{g,\tau}, P_{s,\tau}, \widetilde{C}_{t,\tau})}{\partial C}. \tag{33}$$

Using (25), the derivative of the expenditure function is

$$\begin{aligned} \frac{\partial E(P_g, P_s, C)}{\partial C} = E(P_g, P_s, C)^\sigma \Big[& \omega_c \eta_g P_g^{1-\sigma} C^{(1-\sigma)\eta_g - 1} \\ & + (1-\omega_c) \eta_s P_s^{1-\sigma} C^{(1-\sigma)\eta_s - 1} \Big]. \end{aligned} \tag{34}$$

Therefore, $\widetilde{C}_{t,\tau}$ is characterized by the scalar equation

$$\begin{aligned} \widetilde{C}_{t,\tau}^{-\psi} = \nu_t E(P_{g,\tau}, P_{s,\tau}, \widetilde{C}_{t,\tau})^\sigma \Big[& \omega_c \eta_g P_{g,\tau}^{1-\sigma} \widetilde{C}_{t,\tau}^{(1-\sigma)\eta_g - 1} \\ & + (1-\omega_c) \eta_s P_{s,\tau}^{1-\sigma} \widetilde{C}_{t,\tau}^{(1-\sigma)\eta_s - 1} \Big]. \end{aligned} \tag{35}$$

Given $\widetilde{C}_{t,\tau}$, the hypothetical consumption expenditure is

$$\widetilde{E}_{t,\tau} = E(P_{g,\tau}, P_{s,\tau}, \widetilde{C}_{t,\tau}), \tag{36}$$

and hypothetical investment is

$$\widetilde{x}_{t,\tau} = m_\tau - \widetilde{E}_{t,\tau}. \tag{37}$$

Notice that the optimal hypothetical consumption index $\widetilde{C}_{t,\tau}$, and hence the expenditure level $\widetilde{E}_{t,\tau}$, do not depend directly on income m_τ. This follows from the quasilinear Bellman representation: income enters the problem additively through $\nu_t m_\tau$.

The resulting indirect utility level is

$$\widetilde{V}_{t,\tau} = U(\widetilde{C}_{t,\tau}) + \nu_t \left(m_\tau - \widetilde{E}_{t,\tau} \right). \tag{38}$$

The corresponding hypothetical share of goods in consumption expenditure is

$$\widetilde{s}_{g,t,\tau} = \omega_c P_{g,\tau}^{1-\sigma} \widetilde{C}_{t,\tau}^{(1-\sigma)\eta_g} \widetilde{E}_{t,\tau}^{\sigma-1}, \tag{39}$$

while the hypothetical service share is $\widetilde{s}_{s,t,\tau} = 1 - \widetilde{s}_{g,t,\tau}$.

Second stage. At the second stage, the equivalent variation $\widehat{m}_{t,z,\tau}$ is the income level that, at base prices $(P_{g,z}, P_{s,z})$ and under the same time-t preferences, delivers the utility level $\widetilde{V}_{t,\tau}$ obtained in the first stage.

The household solves

$$\max_{\widehat{C}_{t,z},\, \widehat{x}_{t,z,\tau}} \quad U(\widehat{C}_{t,z}) + \nu_t \widehat{x}_{t,z,\tau} \tag{40}$$

subject to

$$\widehat{m}_{t,z,\tau} = E(P_{g,z}, P_{s,z}, \widehat{C}_{t,z}) + \widehat{x}_{t,z,\tau}. \tag{41}$$

The optimal aggregate consumption index $\widehat{C}_{t,z}$ satisfies

$$\widehat{C}_{t,z}^{-\psi} = \nu_t \frac{\partial E(P_{g,z}, P_{s,z}, \widehat{C}_{t,z})}{\partial C}. \tag{42}$$

Equivalently,

$$\begin{aligned} \widehat{C}_{t,z}^{-\psi} = \nu_t E(P_{g,z}, P_{s,z}, \widehat{C}_{t,z})^\sigma \Big[& \omega_c \eta_g P_{g,z}^{1-\sigma} \widehat{C}_{t,z}^{(1-\sigma)\eta_g - 1} \\ & + (1-\omega_c) \eta_s P_{s,z}^{1-\sigma} \widehat{C}_{t,z}^{(1-\sigma)\eta_s - 1} \Big]. \end{aligned} \tag{43}$$

The associated consumption expenditure is

$$\widehat{E}_{t,z} = E(P_{g,z}, P_{s,z}, \widehat{C}_{t,z}). \tag{44}$$

Importantly, $\widehat{C}_{t,z}$ and $\widehat{E}_{t,z}$ depend on the preference date t and the base-price date z, but not on the comparison period τ.

By construction, $\widehat{m}_{t,z,\tau}$ satisfies

$$U(\widehat{C}_{t,z}) + \nu_t \left(\widehat{m}_{t,z,\tau} - \widehat{E}_{t,z}\right) = \widetilde{V}_{t,\tau}. \tag{45}$$

Using (38), the equivalent variation measure can therefore be written as

$$\begin{aligned}\widehat{m}_{t,z,\tau} &= \widehat{E}_{t,z} + \frac{\widetilde{V}_{t,\tau} - U(\widehat{C}_{t,z})}{\nu_t} \\ &= m_\tau + \frac{U(\widetilde{C}_{t,\tau}) - \nu_t \widetilde{E}_{t,\tau} - \left[U(\widehat{C}_{t,z}) - \nu_t \widehat{E}_{t,z}\right]}{\nu_t}.\end{aligned} \tag{46}$$

It is convenient to define

$$\Phi_t(P_g, P_s) \equiv \max_C \left\{U(C) - \nu_t E(P_g, P_s, C)\right\}. \tag{47}$$

Then the first-stage value can be written as

$$\widetilde{V}_{t,\tau} = \nu_t m_\tau + \Phi_t(P_{g,\tau}, P_{s,\tau}), \tag{48}$$

and the equivalent variation admits the compact representation

$$\boxed{\widehat{m}_{t,z,\tau} = m_\tau + \frac{\Phi_t(P_{g,\tau}, P_{s,\tau}) - \Phi_t(P_{g,z}, P_{s,z})}{\nu_t}}. \tag{49}$$

This expression is the non-homothetic CES counterpart of the closed-form equivalent variation measure obtained under PIGL preferences.

The equivalent-variation measure of the consumption expenditure share is

$$\widehat{s}_{E,t,z,\tau} = \frac{\widehat{E}_{t,z}}{\widehat{m}_{t,z,\tau}}. \tag{50}$$

Since $\widehat{E}_{t,z}$ does not vary with τ, changes in $\widehat{s}_{E,t,z,\tau}$ across comparison periods are driven entirely by changes in the equivalent variation income measure $\widehat{m}_{t,z,\tau}$.

Finally, the equivalent-variation measure of the goods share in consumption expenditure is

$$\widehat{s}_{g,t,z} = \omega_c P_{g,z}^{1-\sigma} \widehat{C}_{t,z}^{(1-\sigma)\eta_g} \widehat{E}_{t,z}^{\sigma-1}. \tag{51}$$

Since both $\widehat{C}_{t,z}$ and $\widehat{E}_{t,z}$ depend only on t and z, the EV goods share is constant across comparison periods:

$$\widehat{s}_{g,t,z,\tau} = \widehat{s}_{g,t,z} \qquad \text{for all } \tau. \tag{52}$$

Thus, as in the benchmark PIGL formulation, the equivalent-variation goods share is invariant to the comparison period τ. Under non-homothetic CES preferences this result follows because, once the preference date t and base-price date z are fixed, the household's Bellman-optimal consumption index and associated expenditure are pinned down independently of τ.